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YUV420 data format

1. YUV Introduction

YUV definition: Is divided into three components, "Y" represents brightness (Luminance or Luma), i.e. the gray value; and "U" and "V" is represented by chromaticity (Chrominance or Chroma), is to describe the image color and saturation, color designation for the pixel.

YUV format： There are two major categories: planar and packed.
For the planar YUV format, into continuous storage of all pixels Y, U subsequently stored in all pixels, followed by all the pixels of the V.
For the packed YUV format, Y of each pixel, U, V * is a continuous cross stored.

YUV storage： Format In fact, its way of sampling is closely related to mainstream sampling in three ways, YUV4: 4: 4, YUV4: 2: 2, YUV4: 2: 0, on the detailed principles, you can learn through online other articles, I would like to emphasize how to restore the samples according to its format YUV values from the code stream of each pixel, because only correctly restore the YUV values for each pixel, to extract the RGB conversion formula of the YUV of each pixel by RGB values, and then displayed.

YUV Features： Is also a method of color coding, luminance information (Y) and color information (UV) separation, no UV information may be displayed as the complete image is only black and white, this design solvesBlack and white TV and color TV compatibility issues. And, YUV not required as three separate RGB video signal transmission simultaneously, so**useYUV mode transmission consumes minimal bandwidth**。

Figure:FIG using three representation intuitively acquired right, black dots represent the Y component of the pixel sampling points, a hollow circle indicates use of the UV component of the pixel.

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2. storage

Now I given in graph form a common stream storage YUV, YUV data and sampling methods are attached to each pixel in the rear storage, wherein, Cb, Cr meaning equivalent to the U, V.

(1) YUVY format (of the YUV422)

YUYV is YUV422A storage format of the sampled adjacent two common adjacent two Y Cb, Cr, analysis for pixel Y'00, Y'01, the value Cb, Cr are Cb00 , Cr00, YUV pixel values of other points and so on.

(2) UYVY format (of the YUV422)

UYVY format is YUV422Storing the sampled format, but is different from the arrangement order YUYV UV just not the same, the reduction method YUV value of each pixel as above.

(3) YUV422P (belonging to the YUV422)

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start + 0:	Y'_{00}	Y'_{01}	Y'_{02}	Y'_{03}
start + 4:	Y'_{10}	Y'_{11}	Y'_{12}	Y'_{13}
start + 8:	Y'_{20}	Y'_{21}	Y'_{22}	Y'_{23}
start + 12:	Y'_{30}	Y'_{31}	Y'_{32}	Y'_{33}
start + 16:	Cb_{00}	Cb_{01}		
start + 18:	Cb_{10}	Cb_{11}		
start + 20:	Cb_{20}	Cb_{21}		
start + 22:	Cb_{30}	Cb_{31}		
start + 24:	Cr_{00}	Cr_{01}		
start + 26:	Cr_{10}	Cr_{11}		
start + 28:	Cr_{20}	Cr_{21}		
start + 30:	Cr_{30}	Cr_{31}		

YUV422P also belong to YUV422One, it is a model Plane, i.e. plane mode, the YUV data is not interleaved storage, but first store all the Y component, and then stores all of the U (Cb) component, and finally store all V (Cr) component, as shown in FIG. YUV values each of which extraction method is to follow the basic pixel extraction YUV422 format, i.e. two Y share one UV. For example, for the pixel Y'00, Y'01, the value Cb, Cr are Cb00, Cr00.

(4) YV12, YU12 format (of the YUV420)

start + 0:	Y'_{00}	Y'_{01}	Y'_{02}	Y'_{03}
start + 4:	Y'_{10}	Y'_{11}	Y'_{12}	Y'_{13}
start + 8:	Y'_{20}	Y'_{21}	Y'_{22}	Y'_{23}
start + 12:	Y'_{30}	Y'_{31}	Y'_{32}	Y'_{33}
start + 16:	Cr_{00}	Cr_{01}		
start + 18:	Cr_{10}	Cr_{11}		
start + 20:	Cb_{00}	Cb_{01}		
start + 22:	Cb_{10}	Cb_{11}		

YU12 and YV12 belong YUV420Format, but also a Plane mode, the Y, U, V components packaged separately, sequentially stored. Which YUV data for each pixel extracting extraction methods follow YUV420 format, i.e. four Y components share a set of UV. Note that the above figure, Y'00, Y'01, Y'10, Y'11 common Cr00, Cb00, and so other.

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(5) NV12, NV21 (belonging to the YUV420)

start + 0:	Y'_{00}	Y'_{01}	Y'_{02}	Y'_{03}
start + 4:	Y'_{10}	Y'_{11}	Y'_{12}	Y'_{13}
start + 8:	Y'_{20}	Y'_{21}	Y'_{22}	Y'_{23}
start + 12:	Y'_{30}	Y'_{31}	Y'_{32}	Y'_{33}
start + 16:	Cb_{00}	Cr_{00}	Cb_{01}	Cr_{01}
start + 20:	Cb_{10}	Cr_{10}	Cb_{11}	Cr_{11}

NV12 and NV21 belong to YUV420 format, Is a two-plane mode, the Y and UV Plane is divided into two, but UV (the CbCr) of interleaved memory, divided into three rather than plane. Extracting a similar manner on, i.e. Y'00, Y'01, Y'10, Y'11 common Cr00, Cb00.

YUV420 planar data storageTo 720 × 488 image size YUV420 planar Example:

Which is stored in the format: Total size (720 × 480 × 3 >> 1) byte,

Divided into three parts: Y component: (720 × 480) bytes; U (Cb) component: (720 × 480 >> 2) bytes; V (Cr) component: (720 × 480 >> 2) bytes

Three inner portion are stored in row-major, between the three parts is Y, U, V sequential storage.

I.e. 0--720 × 480 bytes of data is YUV component value Y, 720 × 480--720 × 480 × 5/4 bytes of U component 720 × 480 × 5/4 --720 × 480 × 3/2 byte is the V component.

4: 2: 2 and 4: 2: 0 conversion:

The easiest way:

YUV4: 2: 2 -> YUV4: 2: 0 Y unchanged the U and V signal values in the row (vertical direction) of a sampled interlaced. YUV4: 2: 0 ---> YUV4: 2: 2 Y constant, each line of the U and V signal values are formed in two successive rows copy data.

In the YUV420, one pixel corresponds to a point Y, a small box 4 * 4 corresponds to a U and V. For all YUV420 images, aligned Y values thereof are identical, since the image is a grayscale image of Y only. YUV420sp YUV420p and UV data format thereof is arranged on a completely different principle. It is a first U 420p after storage, and then stored V, that is to say they are continuous UV. 420sp and it is UV, UV such alternate storage. (See below) with the above

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theory, I can accurately calculate the size of a YUV420 stored in memory. width * hight = Y (sum), U = Y / 4, V = Y / 4.

Therefore, the length of the data in memory YUV420 is width * hight * 3/2,YUV422 data length in the memory is width * hight * 2 。

Suppose a 8X4 resolution YUV image, which shows the format:

Figure: YUV420sp format

Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8
Y9	Y10	Y11	Y12	Y13	Y14	Y15	Y16
Y17	Y18	Y19	Y20	Y21	Y22	Y23	Y24
Y25	Y26	Y27	Y28	Y29	Y30	Y31	Y32
U1	V1	U2	V2	U3	V3	U4	V4
U5	V5	U6	V6	U7	V7	U8	V8

FIG: YUV420p data format of FIG.

Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8
Y9	Y10	Y11	Y12	Y13	Y14	Y15	Y16
Y17	Y18	Y19	Y20	Y21	Y22	Y23	Y24
Y25	Y26	Y27	Y28	Y29	Y30	Y31	Y32
U1	U2	U3	U4	U5	U6	U7	U8
V1	V2	V3	V4	V5	V6	V7	V8

3. YV12 and I420 the difference

In general, direct video data is collected to RGB24 format, a size of a size RGB24 = width × heigth × 3 byte, RGB32 the size = width × heigth × 4, if I420 (i.e. standard format YUV 4: 2 : 0) is the amount of data size = width × heigth × 1.5byte. After data collection RGB24 to, the need for data compression format for the first time. Color space of the image coming from the RGB2YUV. Because, X264 is necessary when performing coding standard YUV (4: 2: 0). But there should be noted that, although also YV12 (4: 2: 0), but YV12 and I420 but it is different,

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some differences in the storage space above. as follows:
After the first data compression RGB24-> YUV (I420). The amount of data will be reduced by half, RGB24-> YUV (YV12) there is a great loss. After X264 encoded, the data amount will be greatly reduced. The encoded data package transmitted in real time through the RTP. On arrival, the data out, is decoded. Upon completion of decoding, data is still YUV format requires a conversion, so the windows drivers can handle, is YUV2RGB24.

YV12: Brightness (rows × columns) + U (rows × columns / 4) + V (rows × columns / 4)
I420: Brightness (rows × columns) + V (rows × columns / 4) + U (rows × columns / 4)
YUY2 is 4: 2: 2 [Y0 U0 Y1 V0]
yuv420p：yyyyyyyy uu vv
yuv420： yuv yuv yuv

YUV420P, Y, U, V components are three-plane format, and I420 is divided into YV12. I420 format (i.e.: YUV), but the opposite is YV12 (i.e.: YVU).
YUV420SP, Y plane format component, UV packaging format, i.e. NV12. Similar NV12 and NV21, U and V are staggered, except that the UV sequence.
I420: YYYYYYYY UU VV =>YUV420P
YV12: YYYYYYYY VV UU =>YUV420P

NV12: YYYYYYYY UV UV =>YUV420SP

NV21: YYYYYYYY VU VU =>YUV420SP

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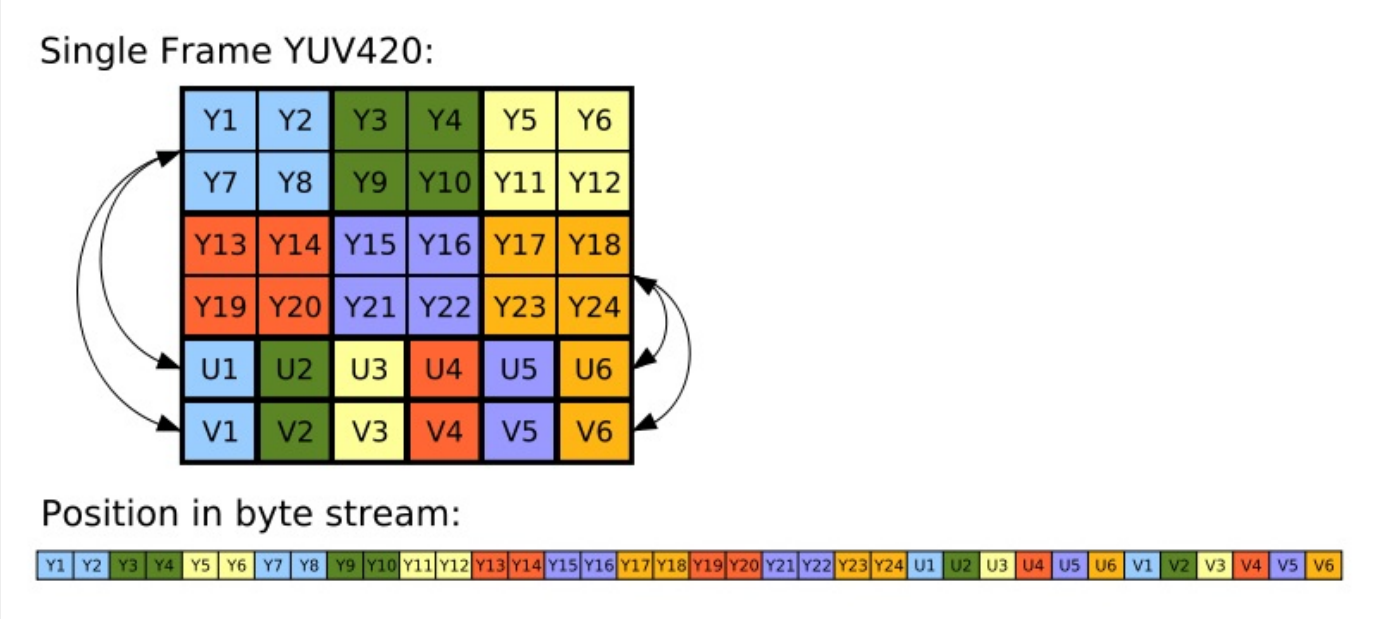

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Intelligent Recommendation

yuv420 data fast cropping

yuv420 data fast cropping I recently encountered a project request. This project has been mass-produced for more than two years. This is a car rearview mirror project. The rearview camera config...

Save YUV420 data to the local

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YUV420 encoding format of AVFrame in FFmpeg

http://m.blog.csdn.net/lanxiaziyi/article/details/74139729#avframe-and-yuv420 those things Using ffmpeg to write the player, originally used sws_scale to convert

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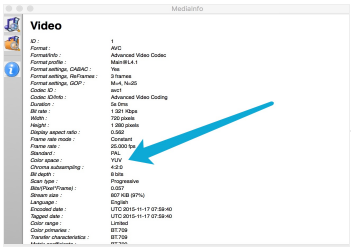
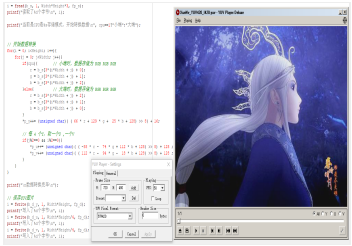


the format after decoding, convert yuv...



[Audio and video data data processing 6] [RGB articles] Convert RGB24 pictures to YUV420 format pictures

[Audio and video data data processing 6] [RGB articles] Convert RGB24 pictures to YUV420 format pictures 1. RGB24 theoretical knowledge 2. YUV420 theoretical knowledge Three, RGB24_to_YUV420_I420 code...

Convert pictures in yuyv422 format to yuv420 format

platform:linux Theoretical basis: In a frame of yuyv422 image, the number of y is the number of pixels, and the number of u and v is half of the number of pixels; in yuyv420, the number of y remains u...

More Recommendation

Display of YUV420 in YV12 format and yuv422 format

During this period of time, I am doing a function of dynamically configuring the character style and color on the video preview frame. Because it needs to be transplanted to several different projects...

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H264 / h265 is transferred to YUV420 data

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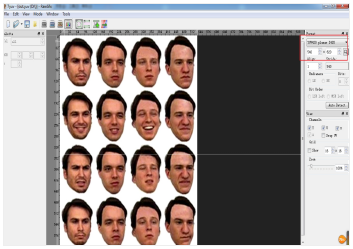
Direct rendering of the YUV420 format using Unity

I haven't been writing a word for a long time. At work, I found that the YUV420 format was converted to RGB format, and finally I still want to render it. So considering direct rendering, saving CPU w...

libwebp encodes yuv420 as webp picture format

yuv420 encoding webp Introduction to libwebp Code Introduction to libwebp Download address and documentation Code...



[OpenCV] Convert image to YUV420 I420 format

One,Introduction to YUV420 I420 A color coding method. In the YUV color space, Y represents the luminance signal, and U and V represent the chrominance signal; The YUV arrangement is as follows, 4 Y c...

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- [Codecs] Detailed graphic explanation of YUV420/YUV422 data format (very comprehensive)
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